REAL-TIME INTRUSION DETECTION SYSTEM

Machine Learning Network Security Report

Generated: 2026-09-21 11:23:48
Reporting Period: 2026-09-09 00:00:00 to 2026-09-21 23:59:59

1. Executive Summary

This report summarizes the network traffic analyzed by the Real-Time Intrusion Detection System. The system uses machine-learning-based classification to identify network flows as either normal traffic or potential intrusions.

Metric	Value
Total Network Flows	3555
Normal Flows	3086
Detected Attacks	469
Attack Rate	13.19%

2. Machine Learning Model

Parameter	Value
Algorithm	Random Forest Classifier
Training Features	duration, protocol_type, src_bytes, dst_bytes, count
Classification	Normal / Attack
Optimized Threshold	0.50
Validation Accuracy	0.76%
Attack Precision	0.96
Attack Recall	0.60
Attack F1-Score	0.74

3. Network Protocol Distribution

Protocol	Flow Count
HTTPS	2051
DNS	557
QUIC_OR_UDP443	534
UDP	235
TCP	127
IPSEC	32
HTTP	18
DHCP	1

4. Detected Attack History

Time	Source	Destination	Protocol	Threat Vector	Severity	Confidence
2026-09-21 11:20:35	10.159.87.202:52675	150.171.110.102:443	HTTPS	ANOMALOUS FLOW	HIGH	98.0%
2026-09-21 11:20:33	10.159.87.123:53	10.159.87.202:59364	DNS	ANOMALOUS FLOW	HIGH	77.86%
2026-09-21 11:20:33	10.159.87.123:53	10.159.87.202:61074	DNS	ANOMALOUS FLOW	HIGH	85.45%
2026-09-21 11:20:29	10.159.87.202:55449	85.210.196.11:443	HTTPS	ANOMALOUS FLOW	HIGH	94.0%
2026-09-21 11:20:23	10.159.87.202:64694	150.171.27.11:443	HTTPS	ANOMALOUS FLOW	HIGH	70.0%
2026-09-21 11:20:15	14.102.231.206:80	10.159.87.202:52259	HTTP	ANOMALOUS FLOW	HIGH	68.0%
2026-09-16 20:44:37	192.168.1.19:59403	10.2.0.1:65432	TCP	ANOMALOUS FLOW	HIGH	94.0%
2026-09-16 20:44:29	192.168.1.19:5353	224.0.0.251:5353	UDP	ANOMALOUS FLOW	HIGH	65.0%
2026-09-16 20:44:15	192.168.1.5:5353	224.0.0.251:5353	UDP	ANOMALOUS FLOW	HIGH	84.42%
2026-09-16 20:43:48	172.64.148.235:443	192.168.1.19:55361	HTTPS	ANOMALOUS FLOW	HIGH	90.0%
2026-09-16 20:43:23	192.168.1.19:59621	172.217.117.4:443	HTTPS	ANOMALOUS FLOW	HIGH	75.2%
2026-09-16 20:43:17	192.168.1.19:60196	4.2.2.2:53	DNS	ANOMALOUS FLOW	HIGH	78.0%
2026-09-16 20:43:16	192.168.1.19:50941	8.8.8.8:53	DNS	ANOMALOUS FLOW	HIGH	100.0%
2026-09-16 20:43:12	192.168.1.19:5353	224.0.0.251:5353	UDP	ANOMALOUS FLOW	HIGH	65.0%
2026-09-16 20:42:44	192.168.1.5:5353	224.0.0.251:5353	UDP	ANOMALOUS FLOW	HIGH	88.42%
2026-09-16 20:42:28	192.168.1.5:5353	224.0.0.251:5353	UDP	ANOMALOUS FLOW	HIGH	84.42%
2026-09-14 18:29:28	192.168.1.10:64764	3.6.199.65:443	HTTPS	ANOMALOUS FLOW	HIGH	90.0%
2026-09-14 18:28:57	104.18.39.21:443	192.168.1.10:57752	HTTPS	ANOMALOUS FLOW	HIGH	94.0%
2026-09-14 18:28:54	3.6.199.65:443	192.168.1.10:64289	HTTPS	ANOMALOUS FLOW	HIGH	100.0%
2026-09-14 18:28:50	192.168.1.10:52100	35.186.224.35:443	HTTPS	ANOMALOUS FLOW	HIGH	98.0%

5. Security Recommendations

1. Investigate repeated attack predictions from the same source address.

2. Monitor network flows with unusually high traffic volume or packet counts.

3. Review suspicious TCP and UDP connections identified by the ML classifier.

4. Use the attack confidence score together with network context when validating alerts.

5. Continue collecting representative network traffic to improve the IDS model.

6. Periodically retrain and evaluate the model using updated datasets.

6. Conclusion

The Real-Time Intrusion Detection System combines live network packet capture, flow-level feature extraction, machine-learning classification, threshold optimization, CSV logging and dashboard-based monitoring. The generated statistics and attack history provide a summary of the traffic observed during the monitoring period.